

Opensourcesoftware Readme

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Open Source Software for Dubbo E-Waste / ITAD

Research into software used by IT asset disposition (ITAD), refurbishment and e-waste operators, with a focus on free/open-source tools and functionality that could be built in-house.

Core finding

A mature ITAD operation usually does not rely on one application. The normal pattern is a combination of:

1. intake and asset/serial tracking
2. chain-of-custody records
3. hardware discovery and diagnostics
4. data sanitisation
5. refurbishment/imaging
6. repair and parts workflows
7. resale/donation/recycling disposition
8. downstream recycler records
9. customer reports and certificates
10. accounting/marketplace integrations

Commercial products such as MAKOR ERP and RazorERP combine many of these functions. A smaller Dubbo operation could begin with open-source specialist tools plus a thin custom orchestration layer rather than attempting to recreate every specialist system.

Strong open-source candidates

Function	Tool	Open source/free	Suggested use
Asset tracking	Snipe-IT	Yes	Asset register, QR/barcodes, status, location and API
ITSM/inventory	GLPI	Yes	Broader service desk + asset system
Automated inventory	OCS Inventory NG	Yes	Discover hardware/software automatically
Disk health	smartmontools	Yes	SMART/NVMe health capture
RAM test	Memtest86+	Yes	Memory validation
Imaging/PXE	FOG Project	Yes	Imaging, PXE, batch refurbishment
Disk wiping	nwipe	Yes	Sanitisation engine
Bootable wipe environment	ShredOS	Yes	Portable nwipe-based wipe station
ERP/business workflow	ERPNext	Yes	Optional accounting/CRM/inventory foundation

Best custom-build opportunities

The strongest areas to build ourselves are the workflow and evidence layer, not the low-level erase algorithms.

Good custom modules include:

- intake forms
- QR/barcode generation and scanning
- chain of custody
- device event timeline
- test result aggregation
- repair/parts workflow
- reuse / donate / sell / recycle routing
- customer/job/project tracking
- downstream recycler tracking
- weight/mass-balance records
- resale and recovery-value tracking
- Certificates of Destruction / Sanitisation / Recycling
- customer portal
- eBay/API integrations
- audit and evidence exports

Important design principle

Do not invent a custom secure-erasure algorithm.

Use established sanitisation software or manufacturer-supported secure erase methods, then build the workflow around them:

```
Asset QR
↓
Drive serial detected
↓
Approved erase method
↓
Erase starts
↓
Verification
↓
Log retained
↓
Operator + timestamp recorded
↓
Certificate generated
```

See:

- software-landscape.md (software-landscape.md)
- assetflow-concept.md (assetflow-concept.md)
- sources.md (sources.md)

Research captured 2 October 2026.

Detailed AssetFlow specification

The vendor reverse-engineering pass added a much more complete target design:

- vendor-workflow-reconstruction.md (vendor-workflow-reconstruction.md) — public MAKOR/RazorERP/Blancco/Phonecheck workflow reconstruction
- assetflow-software-specification.md (assetflow-software-specification.md) — complete functional specification
- screen-map.md (screen-map.md) — proposed operator, admin and client screens
- data-model.md (data-model.md) — conceptual entities and relationships
- feature-backlog.md (feature-backlog.md) — P0-P3 implementation backlog

Important additions from this research

AssetFlow now explicitly includes:

- both bulk lots and serialized assets
- individual media/drive records under parent devices
- lot sorting and mass balance
- configurable workstation/station processing
- customer-specific sanitisation and routing policies
- wipe evidence preservation and hashing
- separate functional/cosmetic grading
- devaluation templates
- automatic/conditional routing
- repair economics
- part harvesting with provenance
- pallets/containers
- outbound/downstream tracking
- settlement and revenue-share workflows
- client Asset Vision / device history
- certificate register and verification
- environmental reporting with explicit methodology
- exception queues
- a visual workflow/rules engine roadmap